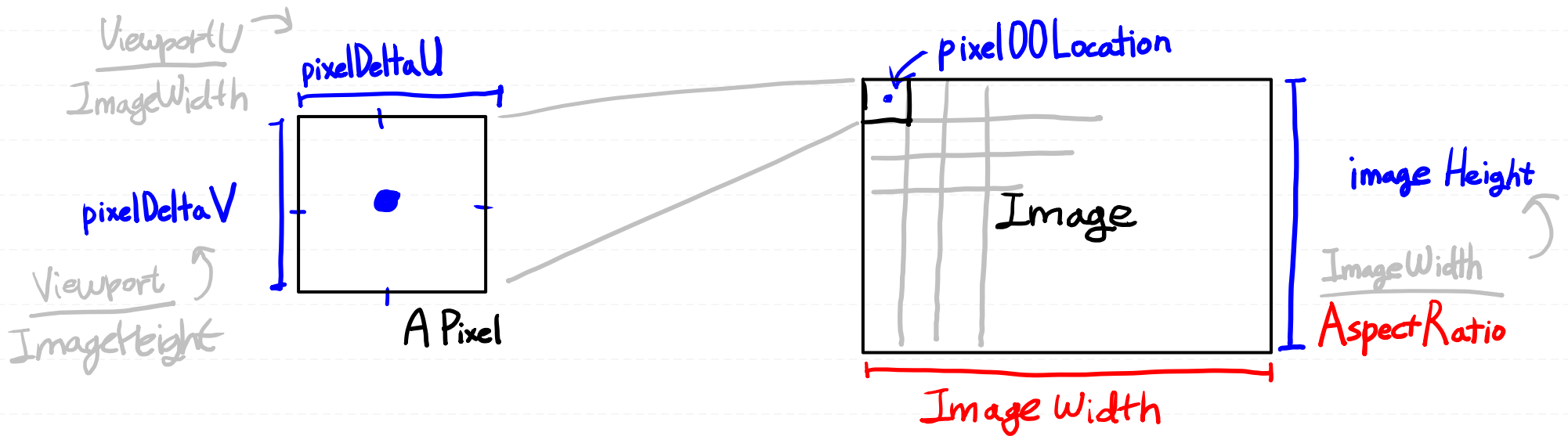
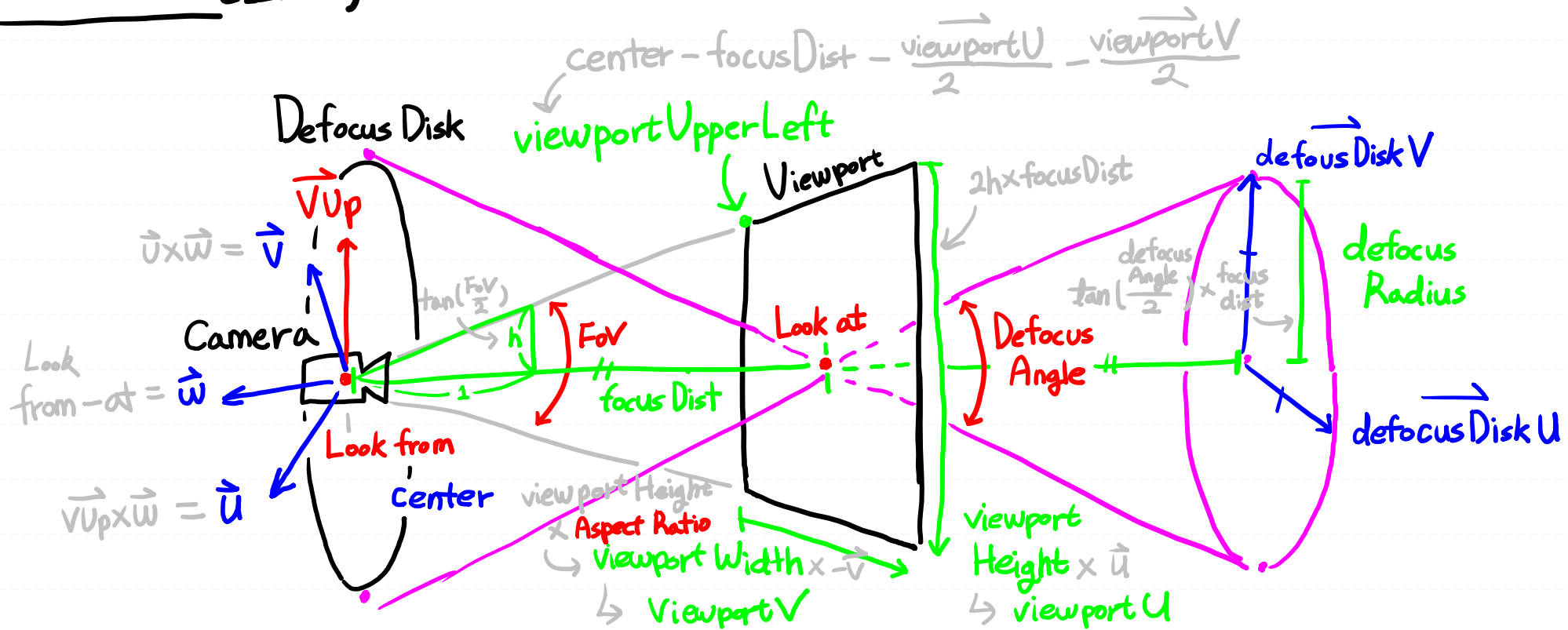
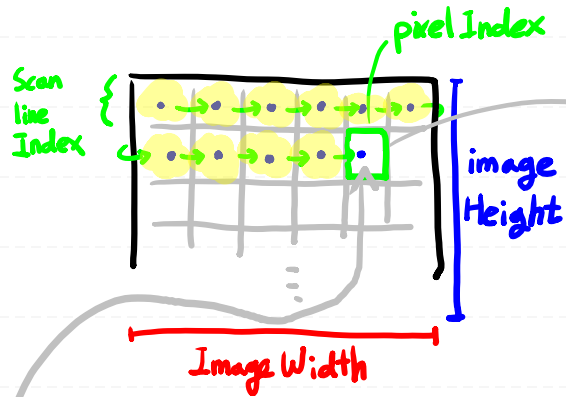


Camera (Init)

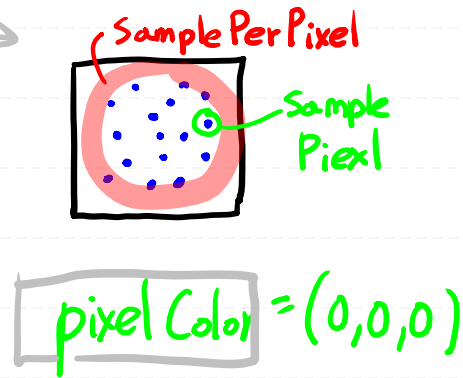


Render

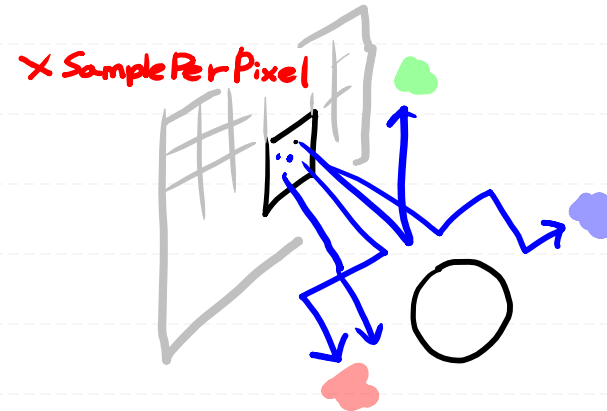
1. Scan Image



2. Sample One Pixel



3. Trace Each Sample



4. Average & Write Color

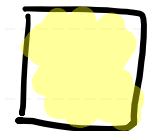
$$\text{Red} + \text{Blue} + \text{Green} + \dots = \text{Orange}$$

$$\text{pixel Color} = C_1 + C_2 + \dots + C_n$$

$$\frac{\text{pixel Color}}{\text{Sample Per Pixel}}$$

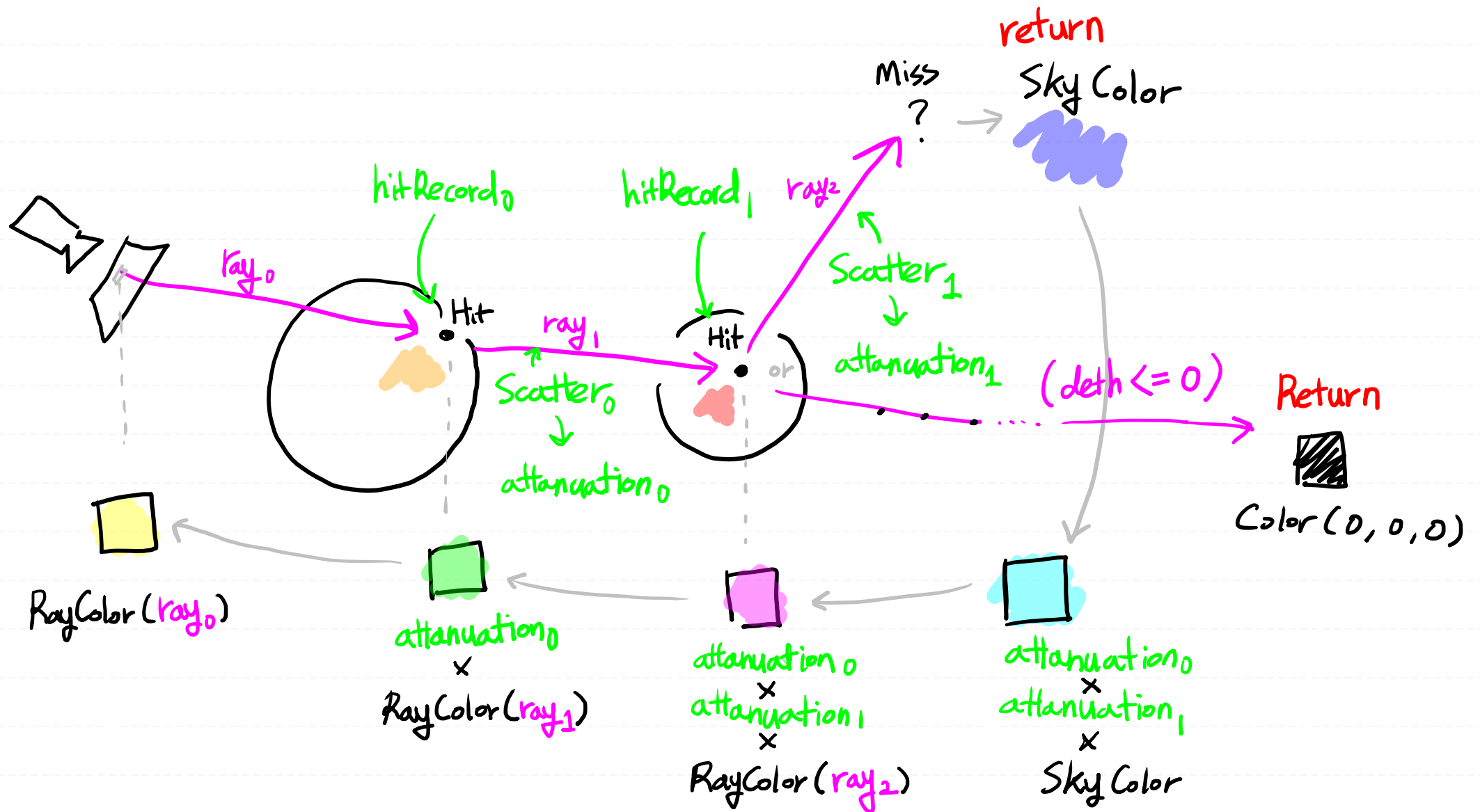
$$\text{Sample Per Pixel}$$

Final Color



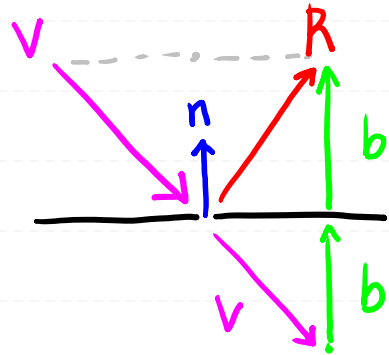
Write Color

Ray Color



Material (Scatter)

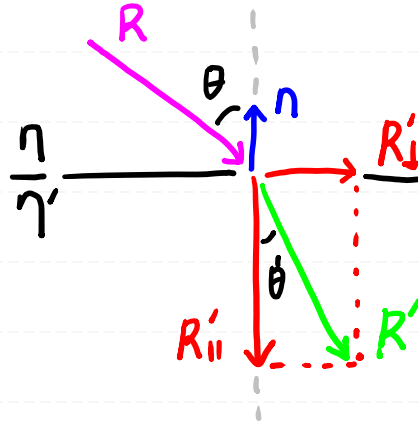
Reflect



$$R = V - 2 \underbrace{\left(\frac{V \cdot n}{|n|} \right)}_b n$$

$\frac{n}{|n|} = n$

Refract



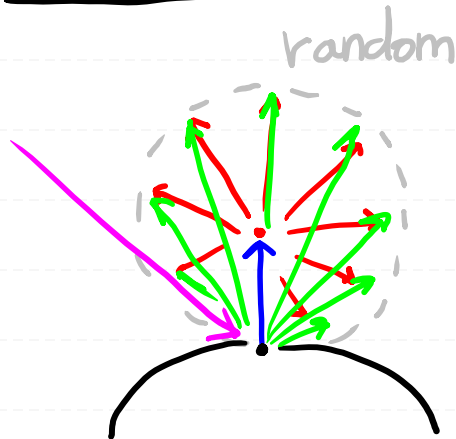
$$|R| \cos \theta = -R \cdot n$$

$$R'_L = \frac{n}{|n|} (R + |R| \cos \theta n)$$

$$R'_{II} = -\sqrt{1 - |R'_L|^2} \cdot n$$

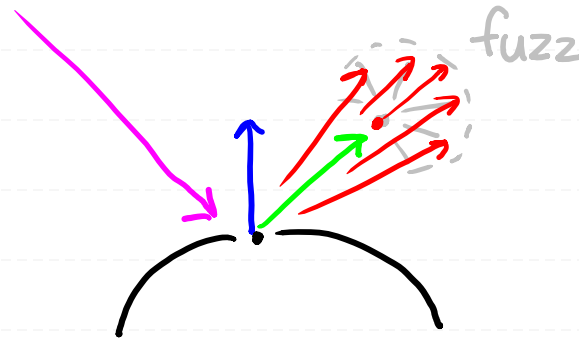
$$R' = R'_L + R'_{II}$$

Lambertian



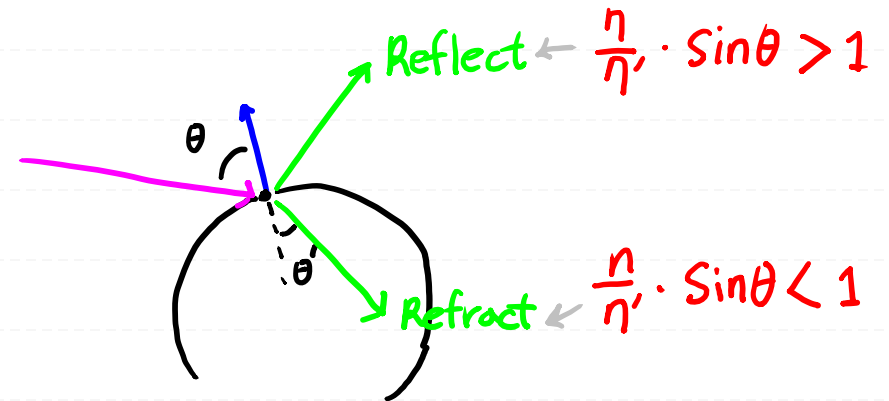
normal + Random
attenuation = albedo

Metal



Reflect + fuzz
attenuation = albedo

Dielectric



Reflect or Refract
attenuation = Color(1,1,1)

$$\text{Reflect} \leftarrow \frac{n}{|n|} \cdot \sin \theta > 1$$

$$\text{Refract} \leftarrow \frac{n}{|n|} \cdot \sin \theta < 1$$